

CHAPTER III.2

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2.3. *Cohomology with supports.* Let X be a topological space, let Y be a closed subset, and let $\mathcal{F}$ be a sheaf of abelian groups. Let $\Gamma_Y(X, \mathcal{F})$ denote the group of sections of $\mathcal{F}$ with support in Y .

(a) Show that $\Gamma_Y(X, \cdot)$ is a left exact functor $\mathcal{A}b(X) \rightarrow \mathcal{A}b$.

We denote the right derived functors of $\Gamma_Y(X, \cdot)$ by $H_Y^i(X, \cdot)$. They are the *cohomology groups of X with supports in Y* , and coefficients in a given sheaf.

(b) If $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ is an exact sequence of sheaves, with $\mathcal{F}'$ flasque, show that

$$0 \rightarrow \Gamma_Y(X, \mathcal{F}') \rightarrow \Gamma_Y(X, \mathcal{F}) \rightarrow \Gamma_Y(X, \mathcal{F}'') \rightarrow 0$$

is exact.

(c) Show that if $\mathcal{F}$ is flasque, then $H_Y^i(X, \mathcal{F}) = 0$ for all $i > 0$.

(d) The sequence

$$0 \rightarrow \Gamma_Y(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{F}) \rightarrow \Gamma(X - Y, \mathcal{F})$$

is exact, with surjectivity of the second map if $\mathcal{F}$ is flasque.

(e) Let $U = X - Y$. Show that for any $\mathcal{F}$, there is a long exact sequence of cohomology groups

$$\begin{aligned} 0 \rightarrow H_Y^0(X, \mathcal{F}) \rightarrow H^0(X, \mathcal{F}) \rightarrow H^0(U, \mathcal{F}|_U) \rightarrow \\ \rightarrow H_Y^1(X, \mathcal{F}) \rightarrow H^1(X, \mathcal{F}) \rightarrow H^1(U, \mathcal{F}|_U) \rightarrow \\ \rightarrow H_Y^2(X, \mathcal{F}) \rightarrow \dots \end{aligned}$$

(f) *Excision.* Let V be an open subset of X containing Y . Then there are natural functorial isomorphisms, for all i and $\mathcal{F}$,

$$H_Y^i(X, \mathcal{F}) \cong H_Y^i(V, \mathcal{F}|_V).$$

Proof. (a) Let $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ be a left exact sequence, and apply $\Gamma(X, \cdot)$ to get a left exact sequence

$$0 \rightarrow \Gamma(X, \mathcal{F}') \xrightarrow{\phi} \Gamma(X, \mathcal{F}) \xrightarrow{\psi} \Gamma(X, \mathcal{F}'')$$

For any $s \in \Gamma(X, \mathcal{F})$ and $p \in X$, $\psi(s)_p = \psi_p(s_p)$. Defining $\psi_Y := \psi|_{\Gamma_Y(X, \mathcal{F})}$, we have $\text{im } \psi_Y \subset \Gamma_Y(X, \mathcal{F}'')$. Defining ϕ_Y similarly, injectivity of each ϕ_p implies $\text{im } \phi_Y = \text{im } \phi \cap \Gamma_Y(X, \mathcal{F}')$. Thus, we can restrict to a sequence

$$0 \rightarrow \Gamma_Y(X, \mathcal{F}') \xrightarrow{\phi_Y} \Gamma_Y(X, \mathcal{F}) \xrightarrow{\psi_Y} \Gamma_Y(X, \mathcal{F}'').$$

Clearly ϕ_Y is still injective. Then

$$\begin{aligned} \ker \psi_Y &= \ker \psi \cap \Gamma_Y(X, \mathcal{F}) \\ &= \text{im } \phi \cap \Gamma_Y(X, \mathcal{F}) \\ &= \text{im } \phi_Y \end{aligned}$$

gives us exactness.

(d) First, take $\mathcal{F}$ arbitrary. Injectivity of the first map is clear. The sheaf axiom gives us that $s|_U = 0$ iff $s_p = 0$ for all $p \in U$, whence we have exactness at the middle term. If we suppose further that $\mathcal{F}$ is flasque, then surjectivity of the second map follows by definition.

(b) (a) and (II, Ex. 1.16b) give us that the following diagram has exact rows, while exactness of the columns follows from (d), using once again that $\mathcal{F}'$ is flasque.

$$\begin{array}{ccccccc}
& & 0 & & 0 & & 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & \Gamma_Y(X, \mathcal{F}') & \xrightarrow{\phi_Y} & \Gamma_Y(X, \mathcal{F}) & \xrightarrow{\psi_Y} & \Gamma_Y(X, \mathcal{F}'') \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & \Gamma(X, \mathcal{F}') & \xrightarrow{\phi} & \Gamma(X, \mathcal{F}) & \xrightarrow{\psi} & \Gamma(X, \mathcal{F}'') \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & \Gamma(U, \mathcal{F}') & \xrightarrow{\phi|_U} & \Gamma(U, \mathcal{F}) & \xrightarrow{\psi|_U} & \Gamma(U, \mathcal{F}'') \longrightarrow 0 \\
& & \downarrow & & & & \\
& & 0 & & & &
\end{array}$$

It suffices to show surjectivity of ψ_Y , which we deduce from the following lemma. It is quite a simple diagram chase, but it is still far clearer to draw out on a whiteboard than to read.

Lemma 1. *If the following commutative diagram of R -modules has exact rows and columns,*

$$\begin{array}{ccccccc}
& & & & 0 & & \\
& & & & \downarrow & & \\
& & A & \xrightarrow{f} & B & & \\
& & \downarrow m & & \downarrow n & & \\
C & \xrightarrow{g} & A' & \xrightarrow{f'} & B' & \longrightarrow & 0 \\
\downarrow p & & \downarrow m' & & \downarrow n' & & \\
C' & \xrightarrow{g'} & A'' & \xrightarrow{f''} & B'' & & \\
\downarrow & & & & & & \\
& & 0 & & & &
\end{array}$$

then $f : A \rightarrow B$ is surjective.

Proof. Fix $b \in B$, and take $a' \in A'$ with $f'a' = nb$. Then $f''m'a' = n'nb = 0$ gives us $c' \in C'$ with $g'c' = m'a'$. Surjectivity of p supplies us with $c \in C$ such that $pc = c'$. Then $m'(a' - gc) = 0$ gives $a \in A$ such that $ma = a' - gc$. Finally, we have

$$nfa = f'ma = f'a' - fgc = nb,$$

so that $fa = b$ by injectivity of n . $\square$

(c) Embed $\mathcal{F}$ in an injective sheaf $\mathcal{I}$, and let $\mathcal{G}$ be the quotient, so that we have an exact sequence

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{I} \rightarrow \mathcal{G} \rightarrow 0.$$

$\mathcal{I}$ is flasque by (2.4), so that $\mathcal{G}$ is flasque as well by (II, 1.16). Taking the long exact sequence of $H_Y^i(X, \cdot)$, (b) gives us $H_Y^1(X, \mathcal{F}) = 0$. Then $H_Y^1(X, \mathcal{I}) = H_Y^1(X, \mathcal{G}) = 0$, so that $H_Y^i(X, \mathcal{F}) = H_Y^i(X, \mathcal{I}) = H_Y^i(X, \mathcal{G}) = 0$ for all $i > 0$ by induction on i .

(e) Take an injective resolution $0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}^\bullet$ of $\mathcal{F}$. Injective sheaves are flasque, so (e) gives us an exact sequence

$$0 \rightarrow \Gamma_Y(X, \mathcal{I}^\bullet) \rightarrow \Gamma(X, \mathcal{I}^\bullet) \rightarrow \Gamma(U, \mathcal{I}^\bullet) \rightarrow 0$$

of complexes. Then, for each $j \geq 0$, we receive the following commutative diagram.

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Gamma_Y(X, \mathcal{I}^j) & \longrightarrow & \Gamma(X, \mathcal{I}^j) & \longrightarrow & \Gamma(U, \mathcal{I}^j) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \Gamma_Y(X, \mathcal{I}^{j+1}) & \longrightarrow & \Gamma(X, \mathcal{I}^{j+1}) & \longrightarrow & \Gamma(U, \mathcal{I}^{j+1}) \longrightarrow 0 \end{array}$$

Applying the snake lemma gives us an exact sequence

$$0 \rightarrow H_Y^i(X, \mathcal{F}) \rightarrow H^i(X, \mathcal{F}) \rightarrow H^i(U, \mathcal{F}) \rightarrow H_Y^{i+1}(X, \mathcal{F}) \rightarrow H^{i+1}(X, \mathcal{F}) \rightarrow H^{i+1}(U, \mathcal{F}),$$

whence we receive our long exact sequence.

(f) Fix a sheaf $\mathcal{F}$, and take $U := X - Y$, so $X = U \cup V$. The sheaf axiom gives us that the map

$$\Gamma(X, \mathcal{F}) \rightarrow \Gamma(V, \mathcal{F}) \oplus \Gamma(U, \mathcal{F})$$

is injective, so the restriction

$$\Gamma_Y(X, \mathcal{F}) \rightarrow \Gamma_Y(V, \mathcal{F}) \oplus \Gamma_Y(U, \mathcal{F})$$

is injective as well. However, $\Gamma_Y(U, \mathcal{F}) = 0$, so the restriction map $\Gamma_Y(X, \mathcal{F}) \rightarrow \Gamma_Y(V, \mathcal{F})$ is injective. If we suppose further that $\mathcal{F}$ is flasque, then it is surjective, so we have an isomorphism

$$\Gamma_Y(X, \mathcal{F}) \xrightarrow{\cong} \Gamma_Y(V, \mathcal{F}) \cong \Gamma_Y(V, \mathcal{F}|_V).$$

Now, take an arbitrary sheaf $\mathcal{F}$, and let $0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}^\bullet$ be an injective resolution. Each $\mathcal{I}^j$ is flasque, so we have an isomorphism

$$\Gamma_Y(X, \mathcal{I}^\bullet) \xrightarrow{\cong} \Gamma_Y(V, \mathcal{I}^\bullet|_V)$$

of complexes. $\mathcal{I}^\bullet|_V$ is a flasque resolution of $\mathcal{F}|_V$, so the homology of the latter still computes $H^\bullet(V, \mathcal{F}|_V)$. Hence, we have an isomorphism

$$H^\bullet(X, \mathcal{F}) \cong H^\bullet(V, \mathcal{F}|_V),$$

as desired. □

2.4. Mayer-Vietoris Sequence. Let Y_1, Y_2 be two closed subsets of X . Then there is a long exact sequence of cohomology with supports

$$\begin{aligned} \cdots \rightarrow H_{Y_1 \cap Y_2}^i(X, \mathcal{F}) &\rightarrow H_{Y_1}^i(X, \mathcal{F}) \oplus H_{Y_2}^i(X, \mathcal{F}) \rightarrow H_{Y_1 \cup Y_2}^i(X, \mathcal{F}) \rightarrow \\ &\rightarrow H_{Y_1 \cap Y_2}^{i+1}(X, \mathcal{F}) \rightarrow \cdots \end{aligned}$$

Proof. It suffices to show that the sequence

$$0 \rightarrow \Gamma_{Y_1 \cap Y_2}(X, \mathcal{F}) \rightarrow \Gamma_{Y_1}(X, \mathcal{F}) \oplus \Gamma_{Y_2}(X, \mathcal{F}) \rightarrow \Gamma_{Y_1 \cup Y_2}(X, \mathcal{F}) \rightarrow 0,$$

with the second map given by $(s_1, s_2) \mapsto s_1 - s_2$, is exact whenever $\mathcal{F}$ is flasque. Exactness in the first two terms is trivial, so it suffices to show surjectivity of the second map. For this, we will once again appeal to the lemma from 2.3(b).

Fix $U_1 := X - Y_1$, $U_2 := X - Y_2$, and consider the following commutative diagram. The columns are exact by flasqueness of $\mathcal{F}$, 2.3(d), and because a component-wise direct sum of two exact sequences is exact.

$$\begin{array}{ccccccc}
& 0 & & 0 & & 0 & \\
& \downarrow & & \downarrow & & \downarrow & \\
0 & \longrightarrow & \Gamma_{Y_1 \cap Y_2}(X, \mathcal{F}) & \longrightarrow & \Gamma_{Y_1}(X, \mathcal{F}) \oplus \Gamma_{Y_2}(X, \mathcal{F}) & \longrightarrow & \Gamma_{Y_1 \cup Y_2}(X, \mathcal{F}) \\
& \downarrow & & \downarrow & & \downarrow & \\
0 & \longrightarrow & \Gamma(X, \mathcal{F}) & \longrightarrow & \Gamma(X, \mathcal{F}) \oplus \Gamma(X, \mathcal{F}) & \longrightarrow & \Gamma(X, \mathcal{F}) \longrightarrow 0 \\
& \downarrow & & \downarrow & & \downarrow & \\
0 & \longrightarrow & \Gamma(U_1 \cup U_2, \mathcal{F}) & \longrightarrow & \Gamma(U_1, \mathcal{F}) \oplus \Gamma(U_2, \mathcal{F}) & \longrightarrow & \Gamma(U_1 \cap U_2, \mathcal{F}) \longrightarrow 0 \\
& \downarrow & & \downarrow & & \downarrow & \\
& 0 & & 0 & & 0 &
\end{array}$$

The top and middles rows are clearly exact. The bottom row is left exact by the sheaf axiom, with surjectivity of the last map following from $\mathcal{F}$ being flasque. Note that exactness of this bottom row, with $U_1 \cup U_2$ taken to be X , gives rise to what is normally referred to as the (non-relative) Mayer-Vietoris sequence in sheaf cohomology.¹

Now, we conclude by Lemma 1 that

$$0 \rightarrow \Gamma_{Y_1 \cap Y_2}(X, \mathcal{F}) \rightarrow \Gamma_{Y_1}(X, \mathcal{F}) \oplus \Gamma_{Y_2}(X, \mathcal{F}) \rightarrow \Gamma_{Y_1 \cup Y_2}(X, \mathcal{F}) \rightarrow 0$$

is exact for $\mathcal{F}$ flasque. Taking an injective resolution $0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}^\bullet$, each $\mathcal{I}^j$ is in particular flasque, so we have an exact sequence of chain complexes

$$0 \rightarrow \Gamma_{Y_1 \cap Y_2}(X, \mathcal{I}^\bullet) \rightarrow \Gamma_{Y_1}(X, \mathcal{I}^\bullet) \oplus \Gamma_{Y_2}(X, \mathcal{I}^\bullet) \rightarrow \Gamma_{Y_1 \cup Y_2}(X, \mathcal{I}^\bullet) \rightarrow 0.$$

Applying the snake lemma as in 2.3(e) gives us our result. $\square$

2.7. (*) Let S^1 be the circle (with its usual topology), and let $\mathbb{Z}$ be the constant sheaf $\mathbb{Z}$.

(a) Show that $H^1(S^1, \mathbb{Z}) \cong \mathbb{Z}$, using our definition of cohomology.

(b) Now let $\mathcal{R}$ be the sheaf of continuous real-valued functions on S^1 . Show that $H^1(S^1, \mathcal{R})$.

Proof. $\square$

2.5. Let X be a Zariski space (II, Ex. 3.17). Let $p \in X$ be a closed point, and let X_p be the subset of X consisting of all points $q \in X$ such that $p \in \{q\}^-$. We call X_p the *local space* of X at p , and give it the induced topology. Let $j : X_p \rightarrow X$ be the inclusion, and for any sheaf $\mathcal{F}$ on X , let $\mathcal{F}_p = j^* \mathcal{F}$. Show that for all i , $\mathcal{F}$, we have

$$H_p^i(X, \mathcal{F}) = H_p^i(X_p, \mathcal{F}_p).$$

¹It seems like Lemma 1 reduces exact sequences in relative cohomology to exact sequences in absolute cohomology, as here and in 2.3(b).

Proof.

□